

Testing Strategy and "Human-in-the-Loop" Requirements

How the System Should Be Tested:

- **Edge-Case Ingestion Testing:** Provide the system with corrupted files, unlisted YouTube videos, and URLs with disabled transcripts to ensure the error-handling and fallback mechanisms trigger correctly without halting batch queues.
- **LLM Hallucination & Accuracy Benchmarking:** Run a curated "golden dataset" of 50-100 known influencer posts through the pipeline. Compare the system's extracted entities, intents, and fact-checking verdicts against the known manual baseline to calculate accuracy and tune the LLM prompts if necessary.
- **Load & Concurrency Testing:** Use the Batch Upload feature to queue a large volume of mixed media files simultaneously to test the ThreadPoolExecutor's efficiency and ensure the SQLite database does not encounter write-lock conflicts under heavy load.

Where the System Still Needs a Human (Human-in-the-Loop): While the automated pipeline processes the bulk of the data, human oversight is critically required in the following areas:

- **Nuance, Sarcasm, and Satire:** The system flags content containing keywords related to sarcasm or satire, but AI notoriously struggles to understand context-heavy irony. A human must review flagged items to determine if the influencer is genuinely endorsing a brand or mocking it.
- **Fact-Checking Resolution:** The fact-checking module acts as a powerful first pass, but it relies on LLM reasoning which can hallucinate. Any claims returned with "Low Confidence" or marked as "Unverified" require a human journalist or researcher to manually verify the claim.
- **Brand Safety & Sensitive Topics:** The system proactively flags "Political Content" and "Sensitive Topics." Because brand safety is highly subjective, a human must review these flagged items to decide if the content poses a PR risk to the campaign.